

```
$ vcsim -h
Usage of vcsim:
  -app int           Number of virtual apps per compute resource
  -cluster int       Number of clusters (default 1)
  -dc int            Number of datacenters (default 1)
  -ds int            Number of local datastores (default 1)
  -esx               Simulate standalone ESX
  -folder int        Number of folders
  -host int          Number of hosts per cluster (default 3)
  -httptest.serve string
                    If non-empty, httptest.NewServer serves on this address and blocks
  -pg int            Number of port groups (default 1)
  -pod int           Number of storage pods per datacenter
  -pool int          Number of resource pools per compute resource
```

Figure 1: Getting help from vcsim

```
$ sudo vcsim -httptest.serve 127.0.0.1:443
GOVC_URL=https://user:pass@127.0.0.1:443/sdk
```

Figure 2: Starting vcsim without any parameters

This should return your home directory path with the Go workspace.

Step 3: Download govcsim using the 'go get' command

```
$ go get github.com/vmware/govmomi/vcsim
$ $GOPATH/bin/vcsim -h
```

If everything is configured correctly, you will be able to get the help options related to govcsim.

To start govcsim without any argument, use the following command:

```
$ vcsim
```

Now, govcsim is working. You can check out the various methods available by visiting <https://127.0.0.1:8989/about> on your favourite browser.

Testing govcsim with Ansible

Now, let's try to write a simple Ansible Playbook, which will list down all VMs emulated by govcsim. The complete code is given in Figure 3. You can read up more about

```
$ cat test_vm_fact.yml
---
- name: Sample Playbook
  hosts: localhost
  vars:
    - vcsim: 127.0.0.1
  tasks:
    - name: List all vms generated using vcsim
      vmware.vmfacts:
        validate_certs: no
        hostname: "{{ vcsim }}"
        username: user
        password: pass
        register: vm_lists
    - debug: msg="{{ vm_lists['virtual_machines'].keys() }}"
$
```

Figure 3: Ansible Playbook to get details about the virtual machine

```
$ ansible-playbook test_vm_fact.yml

PLAY [Sample Playbook] *****
TASK [Gathering Facts] *****
ok: [localhost]

TASK [List all vms generated using vcsim] *****
ok: [localhost]

TASK [debug] *****
ok: [localhost] => {
  "msg": {
    "DC0_C0_IP0_VM0",
    "DC0_C0_IP0_VM1",
    "DC0_H0_VM0",
    "DC0_H0_VM1"
  }
}

PLAY RECAP *****
localhost : ok=3  changed=0  unreachable=0  failed=0

$ _
```

Figure 4: Ansible in action

Ansible at <https://docs.ansible.com/ansible/>.

After running the playbook from Figure 3, you will get a list of virtual machine objects that are simulated by the govcsim server (see Figure 4).

You can play around and write different playbooks to get information about govcsim simulated VMware objects. 

References

- [1] Ansible documentation: <https://docs.ansible.com/ansible>
- [2] Govcsim: <https://github.com/vmware/govmomi/tree/master/vcsim>

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